Maglev Lab Report

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Introduction

Magnetic levitation uses electromagnets to create and control a magnetic field with the goal of levitating a magnetic object. To control the magnetic field created by the electromagnet, a controlled current is fed to the coil in the electromagnet. By controlling this magnetic field, magnetic objects in the field can be manipulated.

The setup for this lab is a magnetic ball levitation device. It consists of an electromagnet, a small magnetic ball, and a position sensor for the ball. This setup is shown in Figure 1.

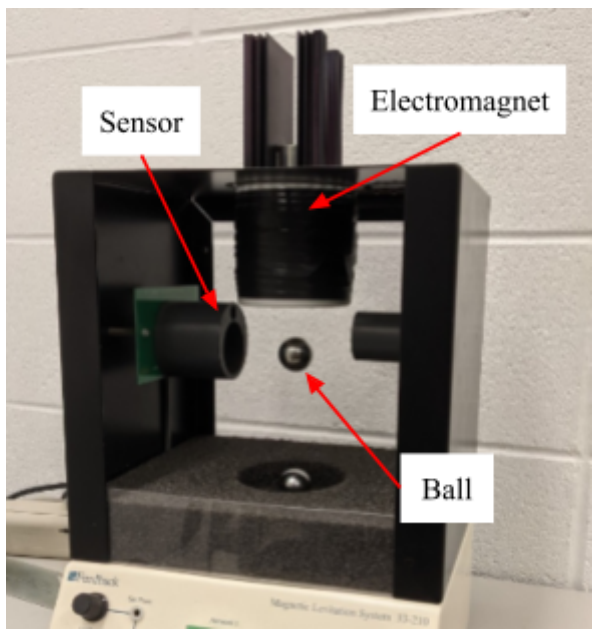

Figure 1: Experimental setup.

Controller Design

In order to control the position of the magnetic ball, a PID controller was designed. This PID controller is based on a linearized model of the maglev system, which contains some nonlinear terms in its equation of motion. The equation of motion is given by equation 1. Equation 2 gives the measurement from the position sensor used the measure the height of the ball

$$m_b \frac{d^2h(t)}{dt^2} = m_b g - \alpha \frac{i(t)^2}{h(t)^2} \tag{1}$$

$$y(t) = \gamma \cdot (h(t) - s_0) \tag{2}$$

In Equation 1, m_b (kg) is the mass of the ball, h(t) (m) is the height of the ball at some time t, g (m/s^2) is the gravitational constant, α (Nm^2/A^2) is the magnetic force constant, and i(t) (A) is the current supplied to the magnet. In Equation 2, y(t) (V) is the position sensor reading, γ (V/m) is the sensor gain, and s_0 (m) is the sensor offset. Table 1 gives the values for each of the constants in the above equations.

Table 1: Constant Values in the Equation of Motion.

Constant	Value
m_b	0.02 kg
g	9.81 m/s^2
α	$2.4832*10^{-5}$ Nm^2/A^2
γ	143.48 V/m
s_0	0.022 m

In this section, we will describe the rationale behind which the controller was designed, including the identification of an equilibrium point and subsequent linearization about this point, before diving into the performance of this controller in simulation.

Equilibrium Point

In order to apply a PID controller, we must first linearize our equation of motion. This requires us to find a point around which we will linearize. At an equilibrium point, unless disturbed, our ball will remain there indefinitely; because of this constant position, all derivatives become zero and greatly simplify our equation of motion. Thus, the team set forth to identify the equilibrium current required to balance the ball at a constant height of h=0.009 m.

Taking Equation 1, we can set all of the derivatives to 0 to find our equilibrium point. Additionally, because the position of the ball, h(t), and current applied to the magnet, i(t), remain constant, they are no longer functions of time; let them now be called $\bar{h}$ and $\bar{i}$, respectively. This gives us Equation 3.

$$0 = m_b g - \alpha \frac{\bar{i}^2}{\bar{h}^2} \tag{3}$$

Solving Equation 3 for $\bar{i}$ yields an equilibrium current of 0.80 A. Thus, our equilibrium point about which we will linearize is $(\bar{h}, \bar{i}) = (0.009\, m,\ 0.80\, A)$. From Equation 2, this corresponds to an equilibrium position reading, $\bar{y}$, of -1.87 V.

Linearization

With an equilibrium point defined, linearization can now be started. Linearization allows us to approximate a nonlinear equation of motion when i(t) and h(t) are close to $\bar{i}$ and $\bar{h}$, respectively. Linearization starts by defining some new variables, the deviations from our equilibrium point. These deviations help shift the equation of motion to be "centered" at our equilibrium point. Each of these deviations is given in Equations 4-7.

$$\delta_h(t) = h(t) - \bar{h} \quad (4) \qquad \frac{d}{dt}(\delta_h(t)) = \frac{d}{dt}(h(t)) \quad (5)$$

$$\frac{d^2}{dt^2}(\delta_h(t)) = \frac{d^2}{dt^2}(h(t)) \quad (6) \qquad \delta_i(t) = i(t) - \bar{i} \quad (7)$$

With the deviations defined, we can move towards calculating the linearized equation of motion. We first rearrange our original equation of motion to the form given by Equation 8, where we call the left-hand-side $f(h(t), i(t))$. From there, Equation 9 is used to calculate the linearized expression of our equation of motion.

$$\frac{d^2}{dt^2}(h(t)) = g - \frac{\alpha}{m_b} \cdot \frac{i(t)^2}{h(t)^2} = f(h(t), i(t)) \quad (8)$$

$$\frac{d^2}{dt^2}(\delta_h(t)) = f(\bar{h}, \bar{i}) + \frac{\delta f}{\delta h}\Big|_{(\bar{h},\bar{i})}\delta_h(t) + \frac{\delta f}{\delta i}\Big|_{(\bar{h},\bar{i})}\delta_i(t) \quad (9)$$

From this linearized system, we generate a transfer function, H(s), from δ_i to δ_h, given in Equation 10. Additionally, we can extend this transfer function to give us G(s), the transfer function from δ_i to δ_y ($\delta_y = y(t) - \bar{y}$), as shown in Equation 11.

$$H(s) = \frac{-24.53}{s^2 - 2180} \quad (10)$$

$$G(s) = \gamma H(s) = \frac{-3519}{s^2 - 2180} \quad (11)$$

With a linearized plant, we are now ready to begin designing the controller for this system.

One Degree of Freedom PID Design

The maglev controller has two requirements for its design: first, it should have zero steady-state error; second, it should have a settling time near 3 seconds. In addition to these requirements, there are some general guidelines to follow in improving the controller, like minimizing the overshoot. The controller designed in this section will be a simple PID controller. Figure 2 shows a block diagram of the controller-plant system.

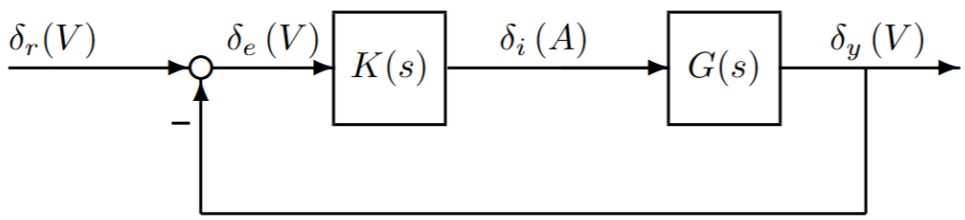

Figure 2: One Degree of Freedom Block Diagram. Note that K(s) is our PID controller.

The first step toward designing the controller was to write out the equation of motion describing the controller-plant system, given by Equation 12. Where a_1 is the coefficient on the s term in Equation 11 (0), a_0 is the coefficient on the s^0 term in Equation 11 (-2180), and b_0 is the numerator coefficient in Equation 11 (-3519). k_d is the derivative gain of our PID controller, k_p is the proportional gain, and k_i is the integral gain. Equation 13 shows the PID controller in transfer function form.

$$\frac{d^3y}{dt^3} + (a_1 + b_0k_d)\frac{d^2y}{dt^2} + (a_0 + b_0k_p)\frac{dy}{dt} + b_ok_iy = b_ok_d\frac{d^2}{dt^2}i + b_ok_p\frac{d}{dt}i + b_0k_ii \qquad (12)$$

$$K(s) = \frac{k_ds^2+k_ps+k_i}{s} \qquad (13)$$

As can be seen, the equation of motion is a third order system, meaning it has 3 poles that we will design for in the pole-placement of our PID controller. With 3 poles, we have some freedom on how to design the controller, but the approach used was to place one "slow" pole at the desired settling time, and two "fast" poles at a higher frequency. Doing this allows us to approximate the system, for design, using the dominant pole approximation, which states the system can largely be described by only the "slowest" pole, provided that other poles are sufficiently fast. Equation 14 gives the form of such a controller-plant system

$$(s^2 + 2\xi\omega_ns + \omega_n^2)(s + p) = s^3 + (p + 2\xi\omega_n)s^2 + (2\xi\omega_np + \omega_n^2)s + \omega_n^2p \qquad (14)$$

Where ξ is the damping ratio and ω_n is the natural frequency of the "fast" poles. The value p determines the location of the "slow" pole that we will use to set the settling time of the system.

Equations 15 and 16 give the settling time for a one-pole and a two-pole system respectively, corresponding to "slow" and "fast" poles above.

$$T_{s-1st\ order} = \frac{3}{|Re(p)|} \tag{15}$$

$$T_{s-2nd\ order} = \frac{3}{\xi\omega_n} \tag{16}$$

Hence, from Equation 15, we set T_{s-1} to the desired 3 second and find that p = +/- 1. We choose p=1 so that the pole is located in the left-half plane. From there, T_{s-2} to be 1/100th of T_{s-1} (0.03 seconds), to ensure that it is sufficiently fast as to not interfere with the dominant pole approximation we are using in design. In addition to the settling time requirements, we also aim to minimize overshoot. This is accomplished with ξ greater than 1 and generally the larger the better (Equation 17). For now, we will let ξ be 2; to get to this value, different values were tested and step responses computed to attempt to minimize the overshoot experienced.

$$\xi \geq 1 \tag{17}$$

With a value of ξ assigned, values for k_d, k_p, and k_i were calculated. To do so, the left-hand side of Equation 12 was set equal to the right-hand side of Equation 14, and the various terms solved for. The values for each of these control parameters is given in Table 2.

Table 2: One Degree of Freedom Control Parameters.

Parameter	Value
k_d	-0.0571
k_p	-1.387
k_i	-0.7105

One Degree of Freedom Linear Simulation Results

To test the function of the PID controller, some simulations were run on the linearized controller-plant system. To do so, the transfer functions from δ_r to δ_y and δ_r to δ_i were generated. Then a step response was generated for a step amplitude of $\delta_r = 1.43\ V$ (roughly a change in ball position of 0.01 m). Figures 3 and 4 show the one degree of freedom response to this step input.

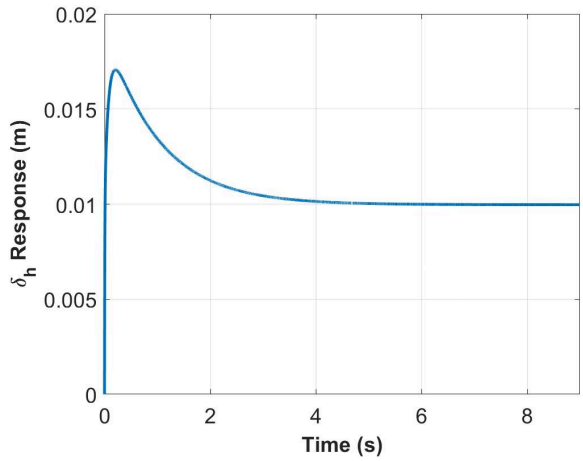

Figure 3: One Degree of Freedom - δ_h step response.

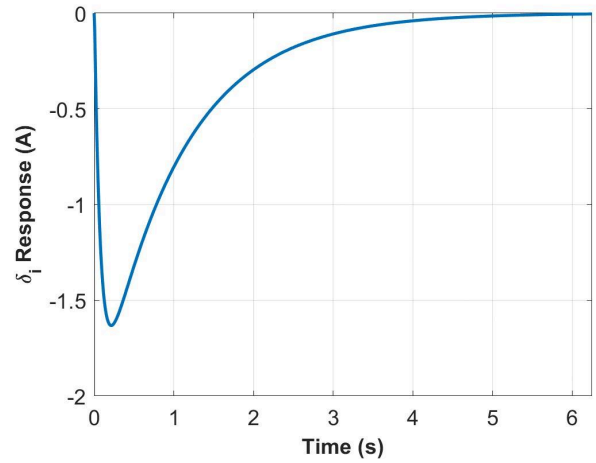

Figure 4: One Degree of Freedom - δ_i step response.

From Figures 3 and 4, we can see that the steady-state behavior of the system is correct, as the position of the ball converges to the reference value supplied. The settling time of the system also appears to be within the 3 second requirement that we have. On the other hand, there is clearly quite a bit of overshoot in the response, not an ideal feature of our controller.

Two Degree of Freedom PID Design

While the one degree of freedom PID design was able to stabilize the plant, it struggled greatly with large overshoot and transient values in its response (δ_y) and control input (δ_i). To help fix this, a reference pre-compensator, F(s), can be used. A pre-compensator helps to smooth the reference command so that it doesn't have large, sudden changes which can contribute to overshoot. The pre-compensator used takes the form of Equation 18 because a PID controller was used. Figure 5 shows a block diagram of the system with the pre-compensator included.

$$F(s) = \frac{k_i}{k_d s^2 + k_p s + k_i} \tag{18}$$

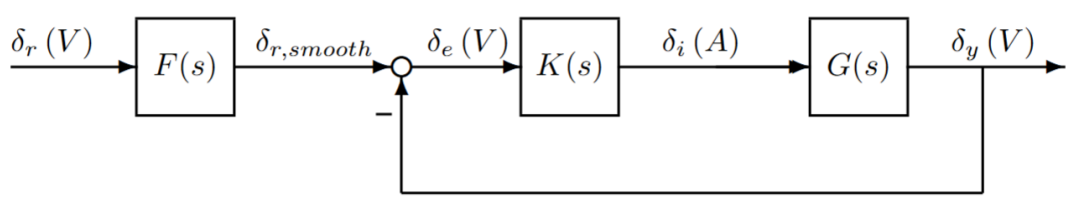

Figure 5: Two Degree of Freedom Block Diagram. Note that K(s) is our PID controller.

Note that the pre-compensator perfectly cancels the zeros of K(s), our PID transfer function. This intentional cancellation leads to less overshoot by eliminating these zeros, which are a main source of overshoot in our controller.

Two Degree of Freedom Linear Simulation Results

To test our controller with the pre-compensator, the same step response from the one degree of freedom simulation was performed. A step response with amplitude of 1.43 V. This resulted in Figures 6 and 7.

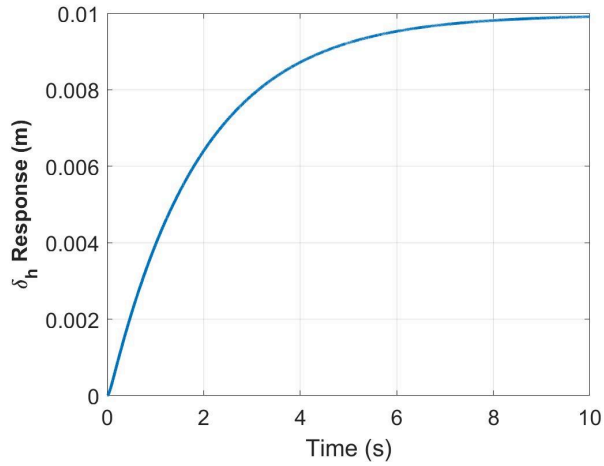

Figure 6: Two Degree of Freedom - δ_h step response.

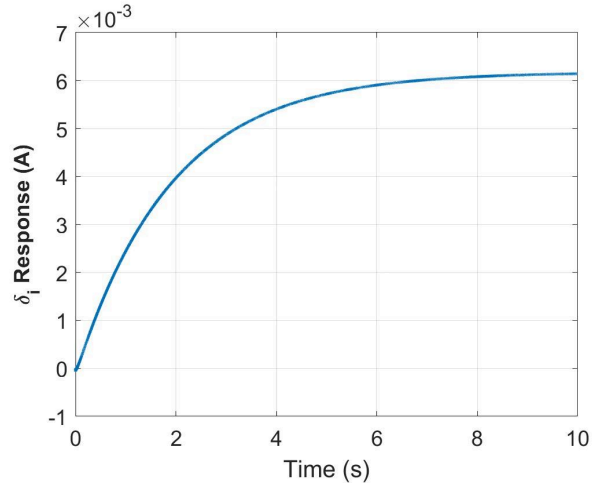

Figure 7: Two Degree of Freedom - δ_i step response.

The responses shown in Figures 6 and 7 clearly do a much better job at reducing the overshoot previously seen in the no pre-compensator responses in Figures 3 and 4. However, the settling time appears to have increased somewhat significantly. Where the settling time previously was ~3 seconds, it is now closer to 5 or 6 seconds, beyond the requirements set forth for this system. To help counteract this, we could go back to our PID design and iteratively improve upon the settling time we see here by changing the location of the lone pole.

In addition to simulating the step responses of our system, stability margins were also calculated to examine how safe our controller is to fluctuations in given parameters. The gain margin examines how much additional gain we can place in our closed-loop system before it becomes unstable. Similarly, the phase margin examines how far our phase is from leading to an unstable system. The delay margin examines how much of a delay the system can experience in controller lag before the system becomes unstable. The gain, phase, and delay margins are given below in Table 3.

Table 3: Controller Stability Parameters.

Parameter	Value
Gain Margin	0.4493
Phase Margin	82.76°
Delay Margin	0.0076 seconds

A good gain margin is generally considered [0.5, 2]; an additional gain can be placed in the closed-loop of between ½ and 2 and the system will still remain stable. Our gain margin is 0.4493, slightly better than the recommended lower gain margin of 0.5, and there is no upper gain margin, which is better than having an upper gain margin of 2. So our system should remain stable for additional gains in the region [0.4493, ∞). A good phase margin is generally considered 45°. The phase margin for our design is 82.76°, well above the recommended value. Finally, a good delay margin is generally considered 2-3 time samples (0.002 - 0.003 seconds for our system with a sampling time of 0.001 seconds). The delay margin for our system, 0.0076 seconds, is again well above the recommended value and should likely cause no issues.

Nonlinear Simulation Results

For the nonlinear simulation, rather than linearizing the plant about some equilibrium point, as was done above, the entire nonlinear plant was used. A step reference was applied with an amplitude of 1.43 V (notably the same as the other step responses) and a step time at 5 seconds. Figures 8 and 9 show the step responses for δ_h and δ_i.

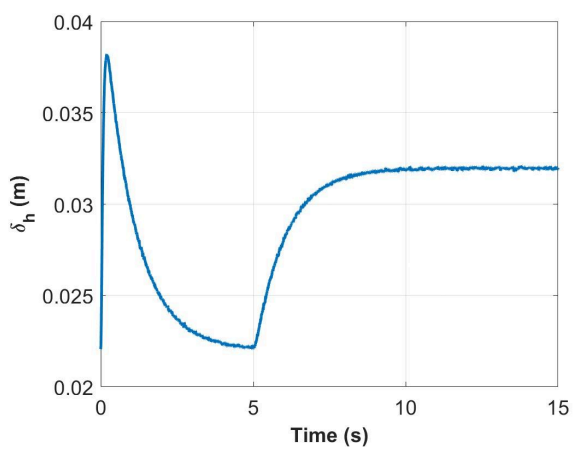

Figure 8: Nonlinear Plant - δ_h step response.

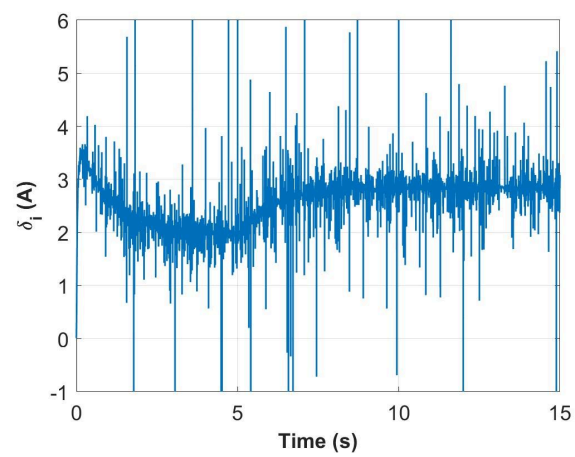

Figure 9: Nonlinear Plant - δ_i step response.

In Figure 8, there is a notable large spike right at the start of the simulation. This spike is caused by the fact that when the simulation first starts at time t=0, the reference position coming through the pre-compensator, $\delta_{r,\,smooth}$, is 0 and remains there until the step time at t=5 seconds. The

output, δ_h, also starts at 0, but quickly begins to increase as gravity acts on the ball; this leads to a very large error between the reference command and the output, which takes the controller a second to catch up to and react accordingly, leading to a very large initial overshoot. To fix this, one would have to "prime" the pre-compensator so that it starts with an initial magnitude to counteract the force of gravity which immediately begins pulling the ball down.

Outside of the initial transient, the δ_h response behaves well and as expected, with minimal error about the steady-state, due only to the sensor noise included in the simulation. The settling time appears to be right around the 3 second requirement, possibly slightly higher but difficult to say for certain. The response for δ_i is a little harder to examine. Due to the relatively high sensor noise in the system, we end up with very large fluctuations and transients in the δ_i response. It still does seem to follow the overall expected shape, and leads to the correct response in δ_h, however, indicating that it is performing well.

Experimental Results

This section will dive into the results on various tests performed on the controller in-action with the maglev system. The experiments performed help to evaluate the stability of our controller and determine whether it meets the specifications of settling time less than 3 seconds and zero steady state error that we defined.

Stabilization

In this section, we evaluated the stability of the closed-loop feedback formed between our controller and the plant. This test is done by providing a constant input to the system and assessing how well the output tracks the reference signal. This test is essential to ensure that the PID controller maintains system stability under steady-state conditions and effectively minimizes steady-state error without introducing excessive oscillations or divergence in the output response.

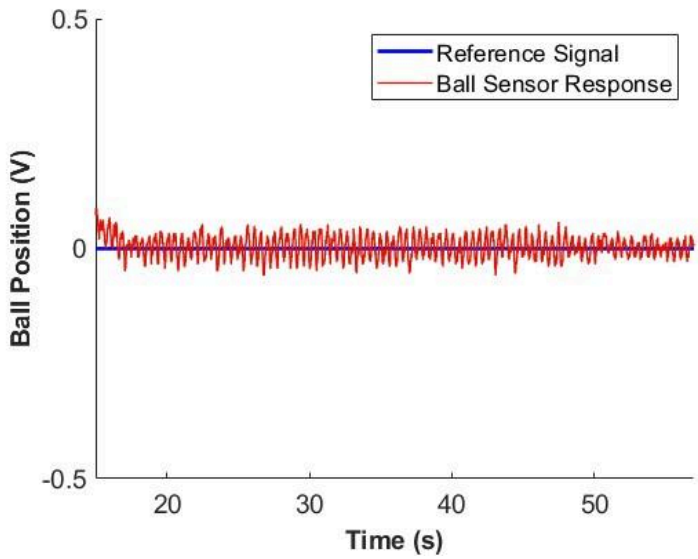

Figure 10: Stability Test Result.

In steady state, the ball position converges to expected reference and stays there, oscillating slightly about its equilibrium position. The average value for y is .003V, which is very slightly higher than the reference signal of 0V, but sufficiently close. No divergence of the reference and output is observed, namely producing 0 steady state error, one of the requirements of our controller. All specifications appear to be met.

Step Response

In this experiment, the system's response was measured in response to a step input. The input was a step of magnitude 0.572 V that occurred at t=25 seconds. This test is crucial to understand how the feedback loop reacts to abrupt changes in input. Therefore providing valuable insights into the sensitivity and response efficiency of the feedback loop.

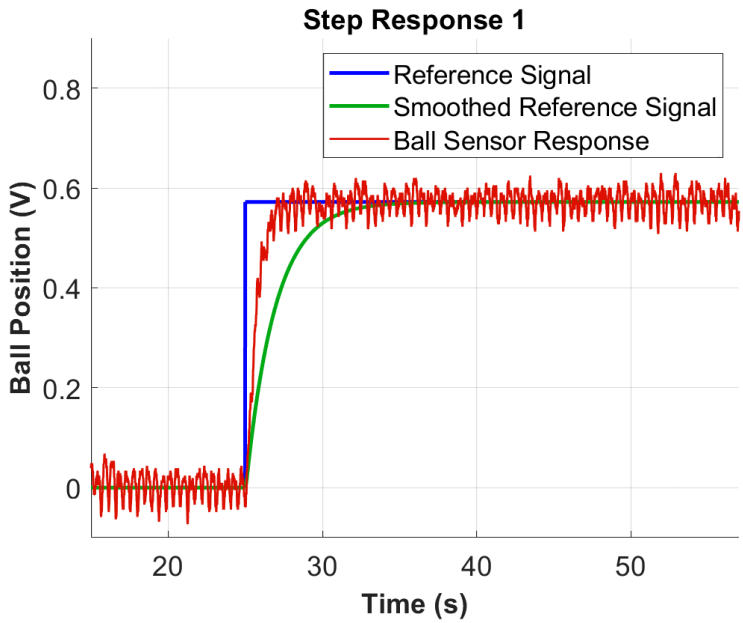

Figure 11: Step Response 1 Test Result. Settling time for the sensor response is approximately 1.94 seconds.

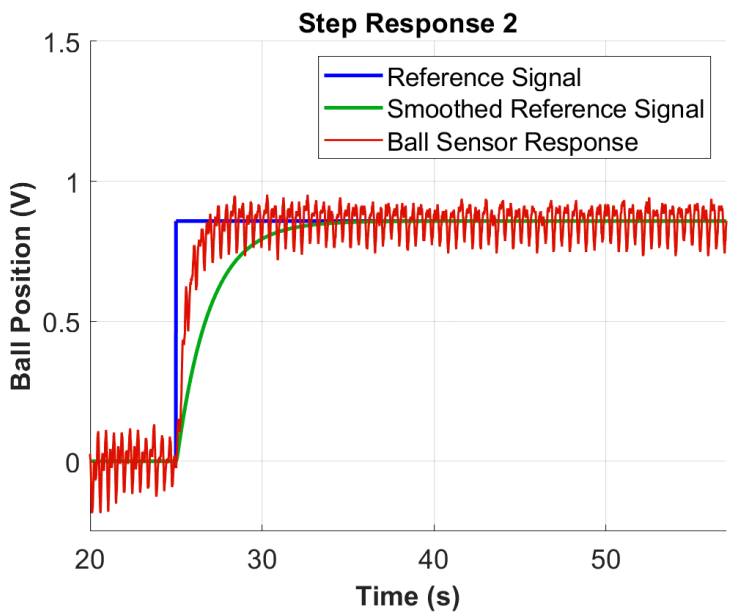

Figure 12: Step Response 2 Test Result. Settling time for the sensor response is approximately 1.48 seconds.

For our evaluation, we suppose settling time to be the time at which the system remains within ±5% of its equilibrium value. We can approximate the settling time by finding the steady-state equilibrium output and looking for the time at which the output arrives at 95% of this value. Note that this is a somewhat imperfect metric for measuring settling time of this response because the steady state behavior does not remain within ±5% of the average value in steady state. Rather, there is enough fluctuation about the average value, that remaining within the 5% error region is

never truly achieved. Nevertheless, following this scheme, we found that the time constant for the ball using the first control is 1.94s. Moreover, the average sensor output at equilibrium is .5725V, which is similar to that of the smoothed reference signal, which is evaluated to be .572V. Hence, the controller meets our specifications of settling time less than 3 seconds and achieves virtually 0 steady state error.

For the second step response, with an amplitude of 0.858 V, the ball sensor response averages at .859V at equilibrium. The settling time is approximated to be 1.48seconds, which is slightly faster than the first step response. The fluctuations about the equilibrium are also noticeably higher than that of the first(the first step response shows a standard deviation of .0227V while the second step response has a standard deviation of .0499V), lower amplitude, step response, due to the increased magnitude of the reference signal.

Disturbance Rejection

To evaluate the system's robustness to external disturbances, a brief and sudden force was applied to the magnetic ball by a gentle poke during steady state operation. The purpose of this test was to assess the controller's ability to reject disturbances and return the ball to its desired equilibrium position with minimal deviation and delay.

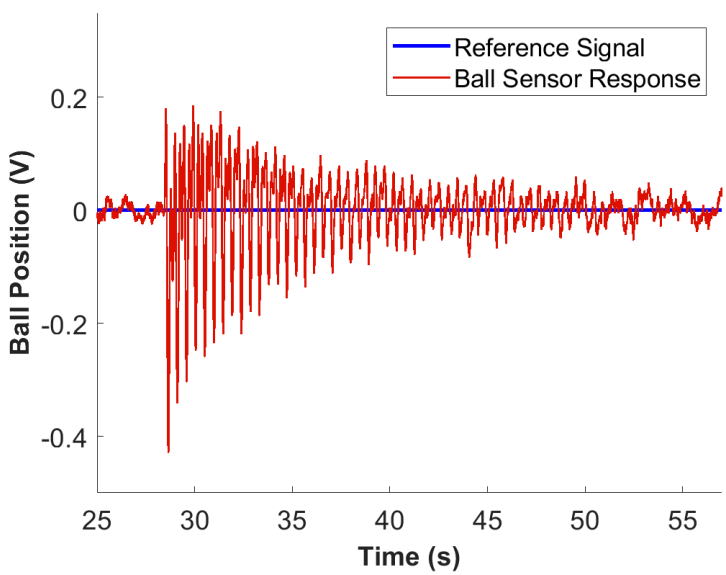

Figure 13: Disturbance Rejection Test Result

As shown in Figure 13, the system responded to the disturbance with a transient deviation in the sensor output, which peaked at approximately 0.4V below the steady state value. However, the controller quickly responded with a response time at 9.8 seconds and the ball's position stabilized, indicating effective disturbance rejection.

No sustained oscillations or instability were observed during the process. The smooth recovery trajectory further confirms that the controller compensates efficiently without inducing additional overshoot or excessive corrective input. Albeit, the response time is slightly slower than the expected 3 second settling time. This supports the conclusion that the closed-loop system maintains robustness under minor external perturbations, a key for reliable levitation.

Staircase Wave Tracking

To assess the controller's dynamic tracking ability, a staircase wave reference signal was applied, consisting of a sequence of step increases in the desired position of the levitating ball. This test is crucial for understanding the system's ability to follow discrete, incremental position changes accurately and with minimal lag or overshoot.

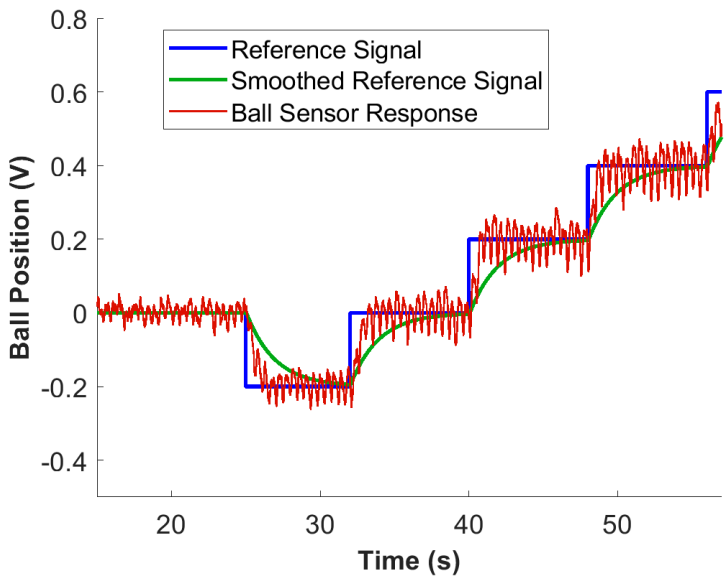

Figure 14: Staircase Wave Tracking Test Result

As shown in Figure 14, the sensor output closely followed each step of the staircase input, demonstrating fast response and accurate tracking at each level change. The rise time for each step was consistently under the 3 second specification, with the system reaching steady state within approximately 1.5s after each transition. Some minor lag was observed during the fastest transitions, but this can be attributed to the physical limitations of the actuator and the presence of sensor noise. Despite these minor delays, the steady state error remained negligible after each step, indicating strong controller performance. Overall, the staircase tracking experiment confirmed the controller's capacity to handle rapid and successive reference updates while maintaining system stability and accuracy.

Square Wave Tracking

To evaluate the controller's performance in tracking a rapidly alternating position, we applied a square wave reference signal with an amplitude of 0.6V. This tests the controller's ability to

respond to abrupt changes in the desired position of the levitating ball and maintain accurate tracking during the rising and falling transitions.

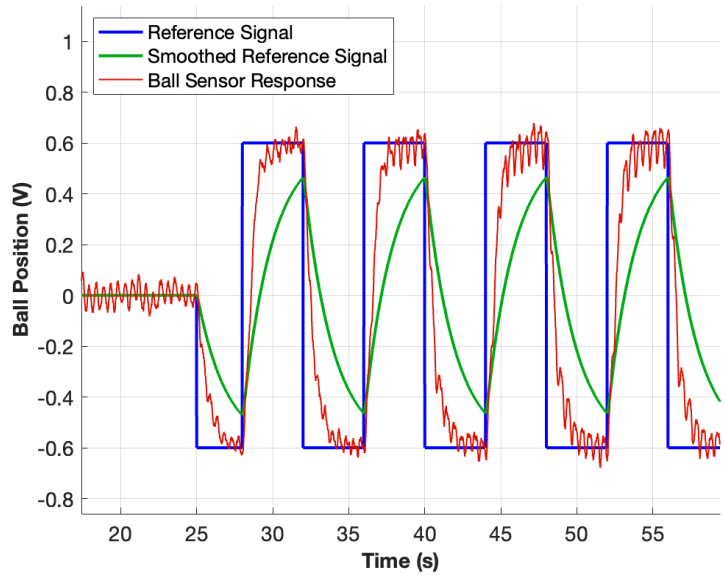

Figure 15: Square Wave Tracking Test Result - 0.6 V Amplitude.

The output closely followed the square wave reference with minimal overshoot, demonstrating the controller's capability to respond to sharp transitions. The settling time is less than 3 seconds which meets our controller specification. During the high and low plateaus of the waveform, the system exhibits stable behavior with low steady state error and the voltage remains about ±0.05V of the reference signal. An additional test was performed with an amplitude of 0.8 as well, shown in Figure 16.

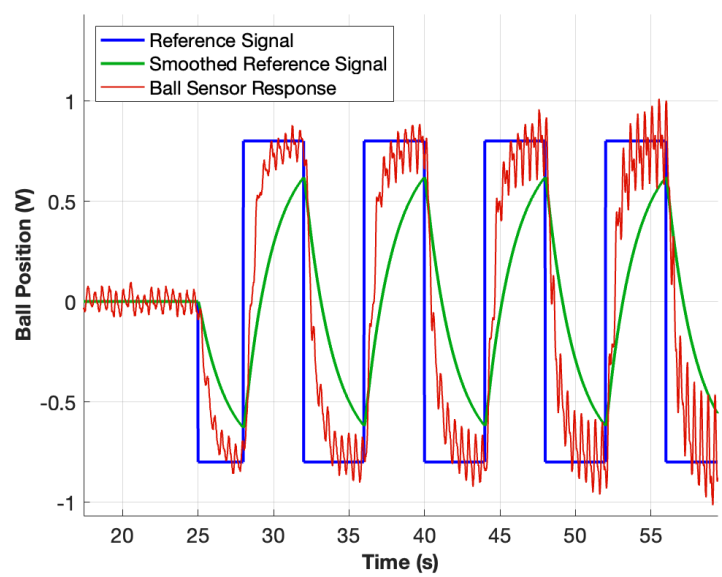

Figure 16: Square Wave Tracking Test Result - 0.8 V Amplitude.

During the first steps, the response was relatively the same, but as the number of steps increased, increased overshoot and oscillations occurred as the test went on. Towards the end the voltage of the ball position had an error of ±0.2V when compared to the reference signal and continued to oscillate more. This large oscillation is bordering on instability and causes insignificant steady-state error. Overall, at lower amplitudes the specifications are met but with amplitude set at higher values, the controller was not able to satisfy all of our requirements.

Conclusion

The magnetic levitation controller designed in this lab met all key performance objectives, both in simulation and on hardware. Using a linearized model of the nonlinear plant, we developed a PID controller with the goal of achieving zero steady-state error and a settling time near 3 seconds. This goal was met through a methodical pole placement approach and further improved via the addition of a reference pre-compensator to reduce overshoot.

The design process proceeded more smoothly than initially anticipated. The linear simulation closely matched expected behavior, and the transition from simulation to real hardware was seamless. The controller was able to stabilize the levitating ball on the very first attempt, indicating a strong match between the theoretical and the physical system.

One unexpected challenge came in managing the trade-offs between overshoot and settling time. While the one degree of freedom controller met the speed requirement, it exhibited significant overshoot. The addition of the pre-compensator in the two degree of freedom architecture effectively reduced the overshoot, though at the cost of increased settling time. Balancing these trade-offs required careful tuning and iterative design, but the final result provided strong performance with robust stability margins.